

MACHINE LEARNING FUNDAMENTALS SERIES

Regularization Part 1: Ridge (L2) Regression

A Mathematical and Conceptual Guide to Shrinkage Penalties, Residuals, and Likelihoods

Source: StatQuest with Josh Starmer — Machine Learning Fundamentals

1 Introduction and Core Motivation

In standard machine learning workflows using linear models (such as Ordinary Least Squares Regression), models are trained by minimizing fitting errors on training data. However, when training sample sizes are small or when datasets contain many predictor variables, standard linear models tend to **overfit** the training data, resulting in high variance and poor generalization to unseen test data [1, 4].

Ridge Regression (L2 Regularization) introduces a small amount of *bias* into parameter estimation in exchange for a substantial drop in *variance*. By shrinking estimated parameters towards zero, Ridge Regression produces models that are less sensitive to noise in training measurements and perform significantly better on new data [4, 7].

2 Residual Calculation and the OLS Baseline

2.1 Definition of Residuals

In regression analysis, a **residual** represents the exact numerical difference between an observed actual value and the model's predicted value for a given sample i [3, 6]:

$$e_i = \text{Residual}_i = y_i - \hat{y}_i = \text{Actual Value}_i - \text{Predicted Value}_i \quad (1)$$

For a simple linear regression model $\hat{y}_i = \beta_0 + \beta_1 x_i$, the residual is $e_i = y_i - (\beta_0 + \beta_1 x_i)$.

2.2 Ordinary Least Squares (OLS) Loss Function

Standard linear regression estimates parameters β_0 (y-intercept) and β_1 (slope) by minimizing the **Sum of Squared Residuals (SSR)** [3, 5]:

$$\text{SSR} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2 \quad (2)$$

2.3 The Small Sample Overfitting Problem

Consider a dataset measuring mouse weight (x) to predict mouse size (y) [2, 4]:

- When evaluated on a large sample, OLS estimates a stable, robust line [3, 4].
- If we extract only $n = 2$ training points (red dots), OLS fits a line passing *exactly* through both points [4].
- Consequently, $\text{SSR}_{\text{train}} = 0$. However, when tested on remaining data (green testing points), the test error SSR_{test} is extremely large [4]. The OLS line exhibits **High Variance** due to an inflated, oversensitive slope [4, 8].

3 Mathematical Mechanism of Ridge Regression

3.1 The Ridge Loss Function

To curb slope inflation, Ridge Regression adds a quadratic penalty term (proportional to the squared slope) to the OLS loss function [5, 6]:

$$\mathcal{L}_{\text{Ridge}}(\beta_0, \beta_1) = \text{SSR} + \lambda \cdot (\text{slope})^2 = \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2 + \lambda \beta_1^2 \quad (3)$$

where $\lambda \geq 0$ is the regularization hyperparameter (tuning parameter) [5, 9].

3.2 Why Ridge Penalizes Steep Slopes (Formula Mechanism)

Why does adding $\lambda \beta_1^2$ prevent overfitting?

1. **Sensitivity to Inputs:** A steep slope ($\beta_1 \gg 0$) implies that small fluctuations in input x cause huge jumps in predicted $\hat{y}$ ($\Delta \hat{y} = \beta_1 \Delta x$) [8]. This makes predictions hypersensitive to training noise [8].
2. **Quadratic Penalty Explosion:** In OLS, increasing β_1 to fit two noisy points reduces SSR to 0. However, in Ridge Regression, as β_1 grows, the penalty term $\lambda \beta_1^2$ grows *quadratically* [5, 6].
3. **Optimal Tradeoff:** Minimizing $\text{SSR} + \lambda \beta_1^2$ forces the optimization algorithm to choose a smaller slope $\beta_1^* < \beta_{1,\text{OLS}}$, trading a tiny increase in training SSR for a dramatic reduction in penalty size and test variance [7, 9].

Numerical Demonstration from StatQuest Video (n = 2 mice)

Let $\lambda = 1$. Consider two candidate line fits [6, 7]:

- **OLS Fit:** Slope $\beta_1 = 1.3$, Residuals = (0, 0) $\implies$ SSR = 0.

$$\text{Loss}_{\text{OLS}} = 0 + 1 \times (1.3)^2 = \mathbf{1.69}$$

- **Ridge Fit:** Slope $\beta_1 = 0.8$, Residuals = (0.3, 0.1) $\implies$ SSR = $0.3^2 + 0.1^2 = 0.10$.

$$\text{Loss}_{\text{Ridge}} = 0.10 + 1 \times (0.8)^2 = 0.10 + 0.64 = \mathbf{0.74}$$

Because $\mathbf{0.74} < \mathbf{1.69}$, Ridge Regression selects the flatter slope ($\beta_1 = 0.8$), reducing prediction sensitivity [7, 8].

4 General Formula for Ridge Regression

4.1 Multiple Linear Regression Formula

For a general dataset with n samples and p predictor variables (features) $x_1, x_2, \dots, x_p$:

$$\hat{y}_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} = \beta_0 + \sum_{j=1}^p \beta_j x_{ij} \quad (4)$$

4.2 General Ridge Loss Function

The general L2-regularized objective function minimizes the sum of squared residuals plus the sum of all squared feature weights multiplied by λ [13, 16]:

$$\mathcal{L}_{\text{Ridge}}(\boldsymbol{\beta}) = \sum_{i=1}^n \left(y_i - \left(\beta_0 + \sum_{j=1}^p \beta_j x_{ij} \right) \right)^2 + \lambda \sum_{j=1}^p \beta_j^2 \quad (5)$$

In vector-matrix notation, letting $\mathbf{y} \in \mathbb{R}^n$, $\mathbf{X} \in \mathbb{R}^{n \times (p+1)}$, and $\boldsymbol{\beta} = [\beta_0, \beta_1, \dots, \beta_p]^T$:

$$\mathcal{L}_{\text{Ridge}}(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda \|\boldsymbol{\beta}_{1:p}\|_2^2 \quad (6)$$

Crucial Rule: Why the Intercept (β_0) is Excluded from the Penalty

Notice that the penalty sum $\sum_{j=1}^p \beta_j^2$ starts at $j = 1$, **excluding the y-intercept** β_0 [13, 16].

Reasoning: β_0 represents the baseline mean of y when all feature inputs $x_j = 0$. Unlike feature weights $\beta_1, \dots, \beta_p$, β_0 is not scaled by sample measurements x_{ij} [13]. Penalizing β_0 would force predictions towards 0 regardless of the natural scale/center of the target variable y [13].

5 Application to Logistic Regression: Sum of Likelihoods

5.1 Why Least Squares is Replaced by Likelihoods

In Logistic Regression, the target $y_i \in \{0, 1\}$ is binary (e.g., Obese vs. Not Obese) [12]. Predictions are modeled via the sigmoid probability function:

$$\hat{p}_i = P(y_i = 1 \mid x_i) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_i)}} \quad (7)$$

Because outcome variables are binary probabilities rather than continuous values, Logistic Regression is solved using **Maximum Likelihood Estimation (MLE)** instead of Least Squares (SSR) [12].

5.2 Sum of the Log-Likelihoods in Ridge Logistic Regression

The likelihood $L(\boldsymbol{\beta})$ measures the joint probability of observing the actual data labels given parameters $\boldsymbol{\beta}$ [12]:

$$L(\boldsymbol{\beta}) = \prod_{i=1}^n \hat{p}_i^{y_i} (1 - \hat{p}_i)^{1-y_i} \quad (8)$$

Taking the natural logarithm yields the **Sum of Log-Likelihoods**:

$$\text{Log-Likelihood}(\boldsymbol{\beta}) = \sum_{i=1}^n [y_i \ln(\hat{p}_i) + (1 - y_i) \ln(1 - \hat{p}_i)] \quad (9)$$

When applying Ridge regularization to Logistic Regression, we maximize the **Sum of Log-Likelihoods minus the L2 Penalty** (or minimize the negative log-likelihood plus penalty) [12]:

$$\text{Objective}_{\text{Ridge Logistic}} = \sum_{i=1}^n [y_i \ln(\hat{p}_i) + (1 - y_i) \ln(1 - \hat{p}_i)] - \lambda \sum_{j=1}^p \beta_j^2 \quad (10)$$

Shrinking β_j prevents logistic probability estimates from becoming overly confident (extreme probabilities near 0 or 1), protecting against classification overfitting [12].

6 Hyperparameter λ and High-Dimensional Problems

6.1 Behavior of Lambda (λ)

- $\lambda = 0$: The penalty term vanishes ($\lambda \sum \beta_j^2 = 0$). Ridge Regression becomes identical to standard OLS / MLE [9, 11].
- $\lambda \rightarrow \infty$: The penalty dominates. To minimize loss, all slopes $\beta_1, \dots, \beta_p$ are forced asymptotically to 0 [10]. The model reduces to a flat horizontal line predicting the sample mean $\hat{y} = \beta_0$ [10].
- **Optimal λ Selection**: Selected experimentally via **10-Fold Cross-Validation** by testing candidate values and picking the λ yielding lowest test variance [10].

6.2 Solving High-Dimensional Problems ($p > n$)

In standard linear regression, analytical parameter estimation requires solving the normal equation:

$$\hat{\beta}_{\text{OLS}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \quad (11)$$

If the number of features p exceeds sample size n (e.g., $p = 10,000$ gene expressions, $n = 500$ mice) [15]:

1. Matrix $\mathbf{X}^T \mathbf{X}$ is singular and non-invertible [14, 15]. OLS has infinitely many solutions with SSR = 0, rendering it unsolvable [14, 15].
2. **Ridge Solution**: Ridge adds $\lambda \mathbf{I}$ to $\mathbf{X}^T \mathbf{X}$, yielding the analytical solution [15, 16]:

$$\hat{\beta}_{\text{Ridge}} = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y} \quad (12)$$

Adding $\lambda \mathbf{I}$ ensures that $(\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})$ is strictly positive-definite and **invertible**, allowing Ridge to uniquely solve high-dimensional problems even when $p \gg n$ [15, 16].